



UNIVERSITY OF
BIRMINGHAM

Superconductivity

Lecture 12 – “Mixed State”

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Previously, supercurrent

Minimise the total free energy F with respect to $\mathbf{A}$,

$$F[\psi, \mathbf{A}] = \int_V d^3\mathbf{r} \left[a|\psi(\mathbf{r})|^2 + \frac{b}{2}|\psi(\mathbf{r})|^4 + \frac{1}{2m^*} |(-i\hbar\nabla + e^*\mathbf{A})\psi(\mathbf{r})|^2 + \frac{B^2}{2\mu_0} \right],$$

to obtain a supercurrent density

$$\mathbf{j}_s = \frac{i\hbar e^{*2}}{2m^*} (\psi^* \nabla \psi - \psi \nabla \psi^*) - \frac{e^{*2}}{m^*} |\psi|^2 \mathbf{A}.$$

Then expand using a complex order parameter,

$$\psi(\mathbf{r}) = |\psi(\mathbf{r})| e^{i\theta(\mathbf{r})},$$

to arrive at

$$\mathbf{j}_s = -\frac{e^*}{m^*} |\psi|^2 (\hbar \nabla \theta + e^* \mathbf{A}).$$

Previously, fluxoid quantisation

Integrate over a contour,

$$m^* \mathbf{v}_s = (\hbar \nabla \theta + e^* \mathbf{A}),$$

to obtain

$$\frac{1}{e^*} \oint_c \frac{m^*}{n^* e^{*2}} \mathbf{j}_s \cdot d\mathbf{r} + \Phi = n \left(\frac{h}{e^*} \right)$$

where $h/e^* \equiv \Phi_0 = h/2e$.

Experimental fluoxid quantisation (1/2)

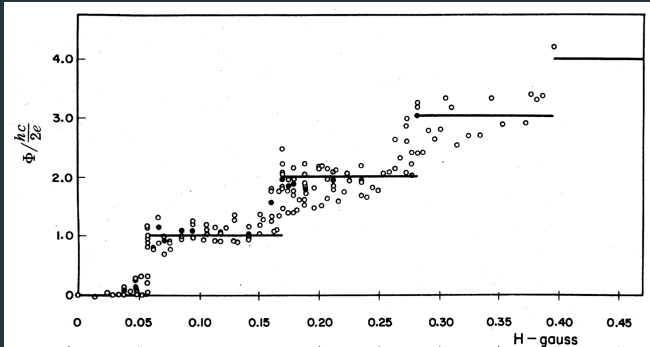


Figure 1: Experiments on Sn cylinder by Deaver and Fairbank.

Experimental fluoxid quantisation (2/2)

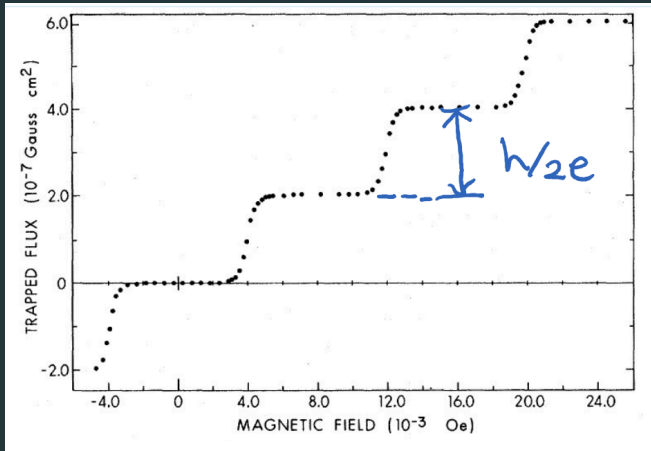


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Vortex

A single vortex

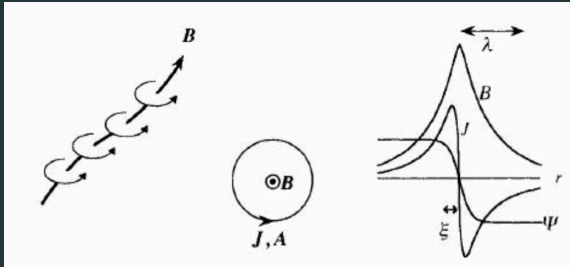


Figure 2: From JR Waldram, Superconductivity of Metals and Cuprates.

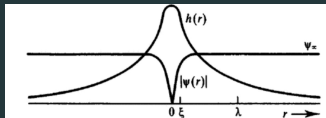


Figure 3: From M Tinkham, Introduction to Superconductivity.

Many vortices

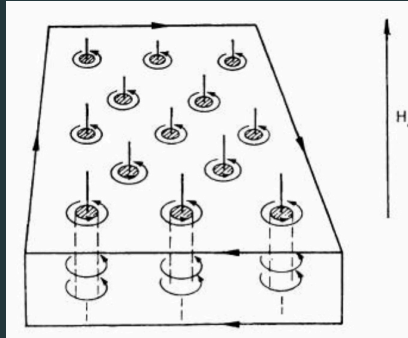
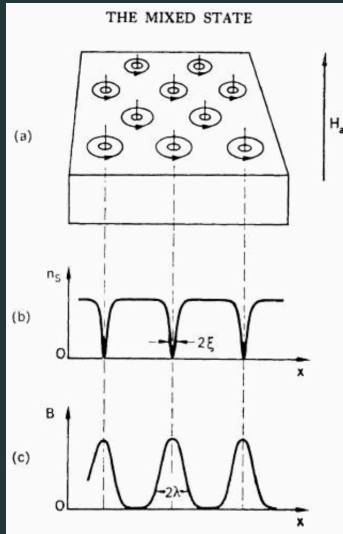


Figure 4: From AC Rose-Innes, Introduction to Superconductivity

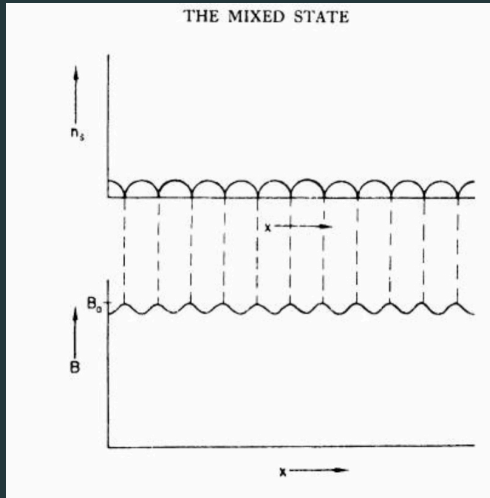
- A vortex experiences the Lorentz force from others;
- Effectively repulsion between vortices.

Many vortices: profile



Many many vortices

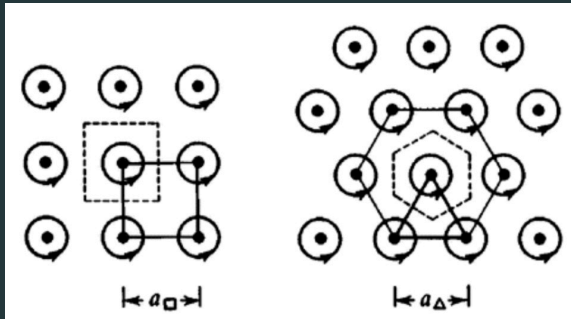
Close to H_{c2}



Flux line lattice

Flux line lattice

Starting from GL equation, one can show that a triangular lattice would form (marginal win over a square lattice).



Flux line lattice: T evolution (from MFM)

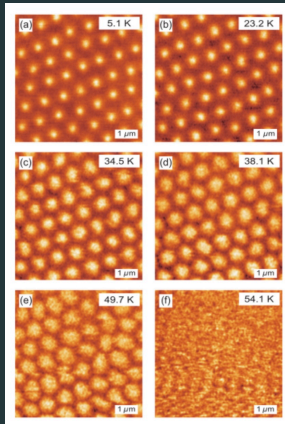


Figure 5: A. Schwarz et al, "Real Space Visualization of Thermal Fluctuations In A Triangular Flux-Line Lattice," New. J. Phys. 12, 033022 (2010).

Flux line lattice: H evolution (from STM)

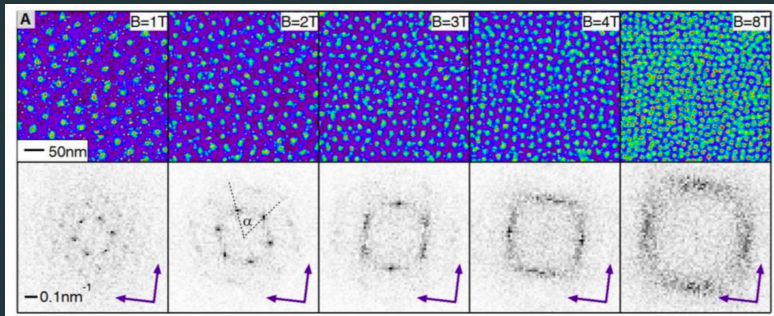
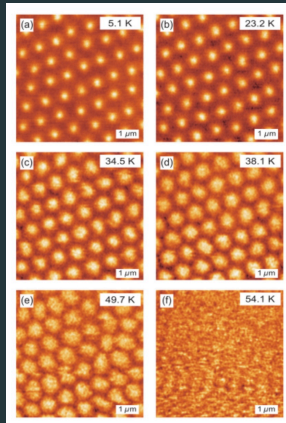


Figure 6: Field evolution of a vortex lattice in LaFeAs superconductor (Ref. J. Yin et al., “Vector field based emergent phase diagram in a correlated superconductor” arXiv 2018).

Flux line lattice: interpretation

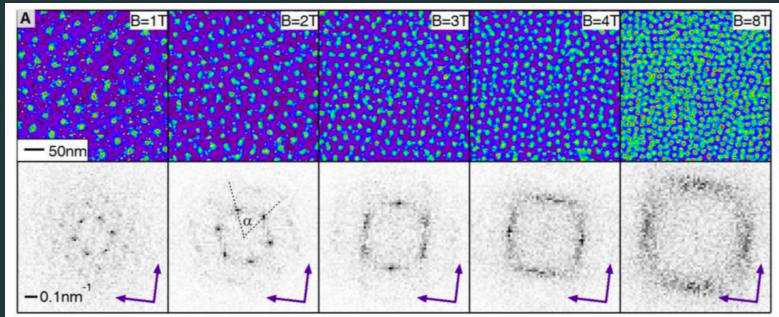
- MFM (Magnetic Force Microscope) uses a magnetic tip.
- STM (Scanning Tunneling Microscope) uses a tunneling current from a tip.
- What exactly are they probing in terms of λ and ξ ?
(What exactly do those spots represent for?)

Flux line lattice: T evolution



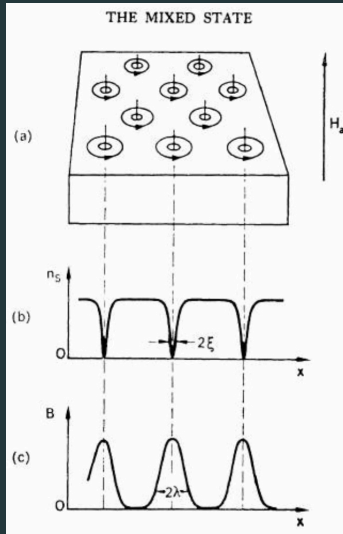
• $\lambda \simeq 100 \text{ nm}$.

Flux line lattice: H evolution



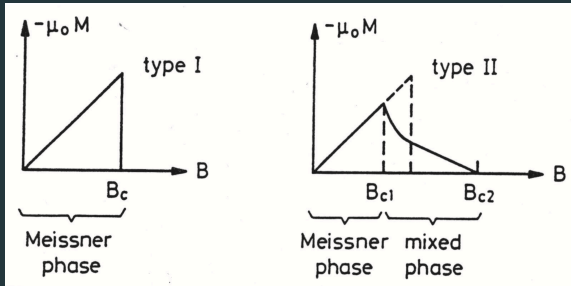
• $\xi \simeq 5\text{ nm}$.

Many vortices: profile



Magnetisation and Currents

Magnetisation



Magnetisation

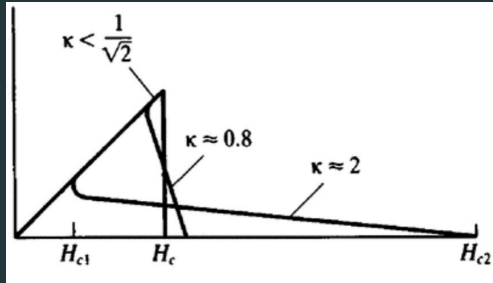


Figure 7: $-M$ versus applied magnetic field.

Critical currents

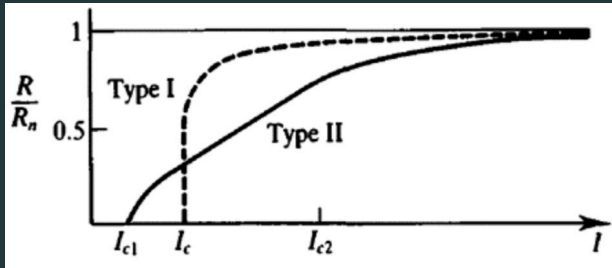
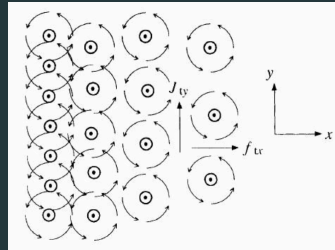
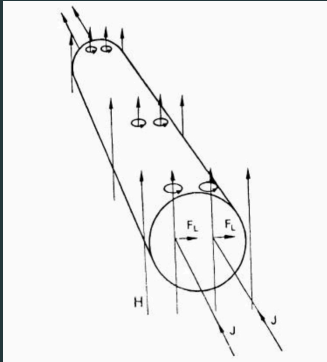


Figure 8: Critical currents for type I and II superconductors, where R is the resistance in superconducting states and R_n in normal states.

Why resistive in mixed states?

Flux line flows



The motion of flux lines

- arises due to Lorentz forces,
- induces resistive voltage and dissipation.

What if we hold the flux lines not to move?

⇒ Introduce defects to trap the flux lines.

Reversible vs. irreversible

